Liangfu Ge

Hong Kong, China liangfu4work@gmail.com civil-liangfu.github.io 0000-0003-0877-1734
Google Scholar Civil-Liangfu

Current Position

Researcher: Department of Civil and Environmental Engineering, The Hong Kong Polytechnic University, Hong Kong, China

Research Interests

- Traffic load identification
- Structural damage detection
- Application of robotics in structural health monitoring (SHM)
- Digital twinning of bridge structures

Methods and Tools

Monitoring and sensing: weigh-in-motion (WIM), machine vision, influence lines, Internet of Things (IoT)

Machine learning: deep learning, object detection (YOLO), transfer learning, domain adaptation, generative adversarial networks, Gaussian process regression

Physics-informed modelling: physics-informed neural networks, Fourier neural operators, vortex-induced vibration, operational modal analysis

Robotics and inspection: ground robots, drones, forward-looking sonar

Publications

VIV-SDE-Net: A physics-informed neural framework for long-horizon prediction of bridge vortex-induced vibrations 2026
Liangfu Ge, Chenqi Wang, Mengliang Liu, Danhui Dan, Fangliang Jian
(Advances in Structural Engineering)

Deep learning-enhanced smart ground robotic system for automated structural damage inspection and mapping 2025
Liangfu Ge, Ayan Sadhu
(Automation in Construction)

Advanced deep learning framework for underwater object detection with multibeam forward-looking sonar 2025
Liangfu Ge, Premjeet Singh, Ayan Sadhu
(Structural Health Monitoring)

A benchmark data set for vision-based traffic load monitoring in a cable-stayed bridge 2024
Liangfu Ge, Danhui Dan, Ayan Sadhu
(Journal of Bridge Engineering)

An improved system for long-term monitoring of full-bridge traffic load distribution on long-span bridges 2023
Liangfu Ge, Danhui Dan, Ki Young Koo, Yifeng Chen
(Structures)

Bridge damage detection using precise vision-based displacement influence lines and weigh-in-motion devices: Experimental validation 2023
Liangfu Ge, Ki Young Koo, Miaomin Wang, James Brownjohn, Danhui Dan
(Engineering Structures)

Intelligent simulation method of bridge traffic flow load combining machine vision and weigh-in-motion monitoring 2022
Liangfu Ge, Danhui Dan, Zijia Liu, Xin Ruan
(IEEE Transactions on Intelligent Transportation Systems)

An accurate and robust monitoring method of full-bridge traffic load distribution based on YOLO-v3 machine vision 2020
Liangfu Ge, Danhui Dan, Hui Li
(Structural Control and Health Monitoring)

Real-time monitoring and evaluation of overturning risk of single-column-pier box-girder bridges based on identification of spatial distribution of moving loads 2020
Liangfu Ge, Danhui Dan, Xingfei Yan, Kailong Zhang
(Engineering Structures)